

Basic Trigonometric #2

$$\cos 120^\circ =$$

$$\tan 150^\circ =$$

$$\sin 150^\circ + \cos 210^\circ =$$

$$\cos 120^\circ$$

$$\begin{aligned} & \swarrow \searrow \\ & \cos(90+30) \quad \cos(180-60) \\ & = -\sin 30^\circ \quad -\cos 60^\circ \\ & = -\frac{1}{2} \quad = -\frac{1}{2} \end{aligned}$$

$$\tan 150^\circ$$

$$\begin{aligned} & \swarrow \searrow \\ & \tan(90+60) \\ & = -\cot 60^\circ \\ & = -\frac{1}{\sqrt{3}} \end{aligned}$$

$$\sin 150^\circ + \cos 210^\circ =$$

$$\sin(180-30) + \cos(180+30)$$

$$\sin 30 - \cos 30$$

$$\frac{1}{2} - \frac{\sqrt{3}}{2}$$

$$1 - \frac{\sqrt{3}}{2} \text{ Ans}$$

ROJGAR WITH ANKIT

Question - $\text{Cosec}^2 30^\circ + \sin^2 45^\circ + \sec^2 60^\circ + \tan^2 30^\circ$ का मान क्या है

$$(2)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + (2)^2 + \left(\frac{1}{\sqrt{3}}\right)^2$$

$$4 + \frac{1}{2} + 4 + \frac{1}{3}$$

$$8 + \frac{1}{2} + \frac{1}{3}$$

$$8 + \frac{5}{6}$$

$$= \frac{53}{6} \text{ Ans}$$

Question $\frac{\sin 30^\circ - \cos 60^\circ + \cot^2 45^\circ}{\cos 30^\circ - \tan 45^\circ + \sin 90^\circ}$ का मान ज्ञात कीजिए के बराबर

$$\frac{\frac{1}{2} - \frac{1}{2} + (1)^2}{\frac{\sqrt{3}}{2} - 1 + 1}$$

$$\frac{\frac{2\sqrt{3}}{3}}{\frac{\sqrt{3}}{2} - 1 + 1}$$

$$\frac{2\sqrt{3}}{\sqrt{3} \times \frac{2}{2}} = \frac{2}{\sqrt{3}} \text{ Ans}$$

$$= \frac{2}{\sqrt{3}}$$

Question Find the value of $(2A + 3B + C)$

$$A = \tan 45^\circ + 3 \cot^2 60^\circ + \sin 30^\circ$$

$$B = \tan^2 30^\circ + \sec 60^\circ + 3 \operatorname{cosec}^2 60^\circ$$

$$C = \operatorname{cosec} 30^\circ + \tan^2 60^\circ + 6 \sec^2 30^\circ$$

$$A = 1 + 3 \times \frac{1}{\sqrt{3}} + \frac{1}{2}$$

$$1 + 3 \times \frac{1}{3} + \frac{1}{2}$$

$$2 + \frac{1}{2} = \frac{5}{2}$$

$$B = \frac{1}{\sqrt{3}} + 2 + 3 \times \left(\frac{2}{\sqrt{3}}\right)^2$$

$$\frac{1}{3} + 2 + 3 \times \frac{4}{3}$$

$$\frac{1}{3} + 6 \rightarrow \frac{19}{3}$$

$$C = 2 + (\sqrt{3})^2 + 6 \times \left(\frac{2}{\sqrt{3}}\right)^2$$

$$2 + 3 + 6 \times \frac{4}{3} = 13$$

$$2 \times \frac{5}{2} + 3 \times \frac{19}{3} + 13$$

$$5 + 19 + 13$$

$$= 37$$

Question — $\frac{4}{3} \tan^2 60^\circ + 3 \cot^2 30^\circ - 2 \sec^2 30^\circ - \frac{3}{4} \csc^2 60^\circ$ is equal to

$$\frac{4}{3} \times \sqrt{3}^2 + 3 \left(\frac{\sqrt{3}}{2} \right)^2 - 2 \times \left(\frac{2}{\sqrt{3}} \right)^2 - \frac{3}{4} \times \left(\frac{1}{\sqrt{3}} \right)^2$$

$$4 + 3 \times \frac{3}{4} - 2 \times \frac{4}{3} - \frac{3}{4} \times \frac{1}{3}$$

$$4 + \frac{9}{4} - \frac{8}{3} - \frac{1}{4}$$

$$\begin{aligned} & 4 + \frac{9}{4} - \frac{8}{3} - \frac{1}{4} \\ & 6 - \frac{8}{3} \\ & = \frac{10}{3} \end{aligned}$$

Question यदि $\sin(A+B) = \cos(A-B) = \frac{\sqrt{3}}{2}$ और A और B न्यूनकोण हैं कोण A और B की माप (डिग्री में) होगी

$$\sin(\overbrace{A+B}^{60^\circ}) = \frac{\sqrt{3}}{2}$$

$$\boxed{A+B = 60^\circ}$$

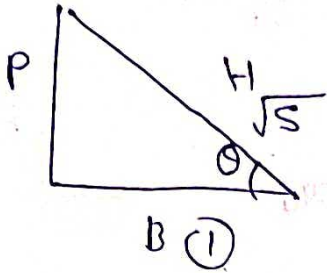
$$\cos(\overbrace{A-B}^{30^\circ}) = \left(\frac{\sqrt{3}}{2} \right)$$

$$\boxed{A-B = 30^\circ}$$

$$\begin{aligned} A &= \frac{60+30}{2} \\ &= 45 \end{aligned}$$

Question यदि $\cos \theta = \frac{1}{\sqrt{5}}$ जहाँ $0 < \theta < \pi/2$ तो

$\frac{2 \tan \theta}{1 - \tan^2 \theta}$ के मान है



$$\cos \theta = \frac{B}{H} = \frac{1}{\sqrt{5}}$$

$$P^2 + B^2 = H^2$$

$$P^2 + 1^2 = (\sqrt{5})^2$$

$$P^2 =$$

1. If $\cos x = \frac{24}{25}$, $0 \leq x \leq 90^\circ$ then what is the value of $\cot x + \operatorname{cosec} x$?

यदि $\cos x = \frac{24}{25}$, $0 \leq x \leq 90^\circ$ है, तो $\cot x + \operatorname{cosec} x$ का मान क्या होगा?

- (a) 7
- (b) 0
- (c) $7/2$
- (d) 1

2. If $\sec \theta = \frac{13}{5}$, then find the value of $\frac{10 \tan \theta + 24 \operatorname{cosec} \theta}{39 \sin \theta - 10 \sec \theta}$.

अगर $\sec \theta = \frac{13}{5}$ है, तो $\frac{10 \tan \theta + 24 \operatorname{cosec} \theta}{39 \sin \theta - 10 \sec \theta}$ का मान ज्ञात कीजिए

- (a) 2
- (b) $1/5$
- (c) 3
- (d) 5

3. If $\sec A = \frac{5}{3}$, then find the value of $\cot A$.

यदि $\sec A = \frac{5}{3}$ है, तो $\cot A$ का मान ज्ञात कीजिए

- (a) $\frac{3}{4}$
- (b) $\frac{3}{5}$
- (c) $\frac{4}{5}$
- (d) $\frac{4}{3}$

4. Find the value of $\cos 0^\circ \cos 30^\circ \cos 45^\circ \cos 60^\circ \cos 90^\circ$.

$\cos 0^\circ \cos 30^\circ \cos 45^\circ \cos 60^\circ \cos 90^\circ$ मान ज्ञात कीजिए

- (a) 3
- (b) $\frac{\sqrt{6}}{8}$
- (c) 0
- (d) 5

5. In $\triangle ABC$, B is right angled, if $\cot A = \frac{1}{2}$, then find the value of $\frac{\sin A(\cos C + \cos A)}{\cos C(\sin C - \sin A)}$

$\triangle ABC$ में, B समकोण हैं, यदि $\cot A = \frac{1}{2}$ है, तो $\frac{\sin A(\cos C + \cos A)}{\cos C(\sin C - \sin A)}$ का मान ज्ञात करें।

(a) 3 (b) -3

(c) 2 (d) -2

6. If $\tan x = \frac{3}{2}$, then what is the value of $\frac{3\sin x + 2\cos x}{3\sin x - 2\cos x}$.

यदि $\tan x = \frac{3}{2}$ है, तो $\frac{3\sin x + 2\cos x}{3\sin x - 2\cos x}$ का मान क्या होगा?

(a) 13/5 (b) 5

(c) 5/13 (d) 1/5

7. What is the value of $(\sin^4 45^\circ + \cos^4 60^\circ) + (\tan^4 45^\circ + \cot^4 45^\circ)$?

$(\sin^4 45^\circ + \cos^4 60^\circ) + (\tan^4 45^\circ + \cot^4 45^\circ)$ का मान क्या है?

(a) 37/16 (b) 33/16

(c) 35/16 (d) 39/16

8. What is the value of $\sin A = \frac{1}{2}(\tan A + \cos A)$?

यदि $\sin A = \frac{1}{2}(\tan A + \cos A)$ का मान क्या है?

(a) $\frac{2}{3\sqrt{3}}$

(b) $\frac{3}{2\sqrt{3}}$

(c) $\frac{5}{2\sqrt{3}}$

(d) $\frac{5}{3\sqrt{3}}$

9. If $\cos \theta = \frac{5}{13}$, then find the value of $\tan^2 \theta + \sec^2 \theta$.

यदि $\cos \theta = \frac{5}{13}$ है, तो $\tan^2 \theta + \sec^2 \theta$ का मान ज्ञात कीजिए।

(a) 233/25

(b) 303/25

(c) 313/25

(d) 323/25

10. Find the value of $\sqrt{\tan^2 60^\circ + \sin 90^\circ - 2\tan 45^\circ}$

$\sqrt{\tan^2 60^\circ + \sin 90^\circ - 2\tan 45^\circ}$ का मान ज्ञात कीजिए

(a) 0

(b) 4

(c) 1

(d) 2

11. If $\tan \theta = \frac{3}{5}$, ($0^\circ < \theta < 90^\circ$) then $\sin \theta \cdot \cos \theta$ is equal to?

यदि $\tan \theta = \frac{3}{5}$, ($0^\circ < \theta < 90^\circ$), तो $\sin \theta \cdot \cos \theta$ किसके समान होगा?

(a) 15/34

(b) $14/\sqrt{34}$

(c) $\sqrt{17}$

(d) $16/\sqrt{34}$

12. Find the value of $\frac{\sin^2 30^\circ + \cos^2 60^\circ + \sec 45^\circ \cdot \sin 45^\circ}{\sec 60^\circ + \operatorname{cosec} 30^\circ}$

$\frac{\sin^2 30^\circ + \cos^2 60^\circ + \sec 45^\circ \cdot \sin 45^\circ}{\sec 60^\circ + \operatorname{cosec} 30^\circ}$ का मान ज्ञात करें।

(a) $-3/8$

(b) $1/4$

(c) $-1/4$

(d) $3/8$

ANSWER SHEET

1	2	3	4	5	6	7	8	9	10
A	D	A	C	B	A	A	C	C	A
11	12								
A	D								